

Sri Lanka Institute of Information Technology

Data Warehousing and Business Intelligence

Assignment 1 – 2025, Semester

Documentation



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Dataset

This research data utilized its dataset from an example set of supermarket sales used commercially. Each of such rows of the used dataset by this research to design their sample were designed similar to Online Transaction Processing and reported every retail scenario under which individuals purchased goods from a storekeeper. For simulating the actual atmosphere of a supermarket, the utilized dataset in this research was tailored-made. A full year's worth of information is addressed, such as transaction history, customer data, product details, and transaction completion times.

For business reporting and intelligence purposes, this transactional information will need to be analyzed and sorted into a dimensional data warehouse.

To provide time-based reporting coverage, an additional year's worth of data has been generated..

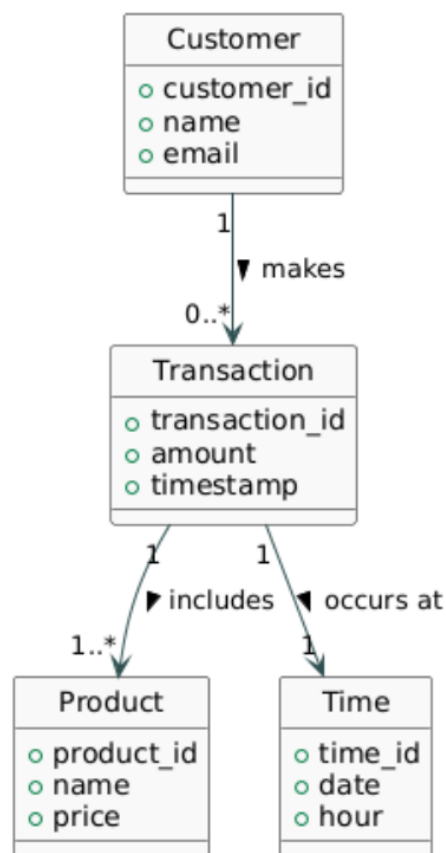
The dataset consists of the following tables

- Customer - Contains customer information such as CustomerID, Full Name, Gender, Email, Age, and Country
- Product – contains all details regarding ProductID, ProductName, Category, and UnitPrice.
- Transaction - contains all details regarding TransactionID, ProductID, CustomerID, Quantity, and TransactionDate.
- Sale - This is my fact table (Fact_Transactions) in the data warehouse, combining keys and metrics for analysis

These files were initially provided in the following formats

- **Customers.csv** – comma-separated values file containing raw customer data
- **Products.xlsx** – Excel file with product details
- **Transactions.sql** – Pre-loaded table in SQL Server with historical sales transactions
- **Complete_times.csv** – Used to update completion timestamps for each transaction

Diagram To show the relationship



Preparation of Data Sources

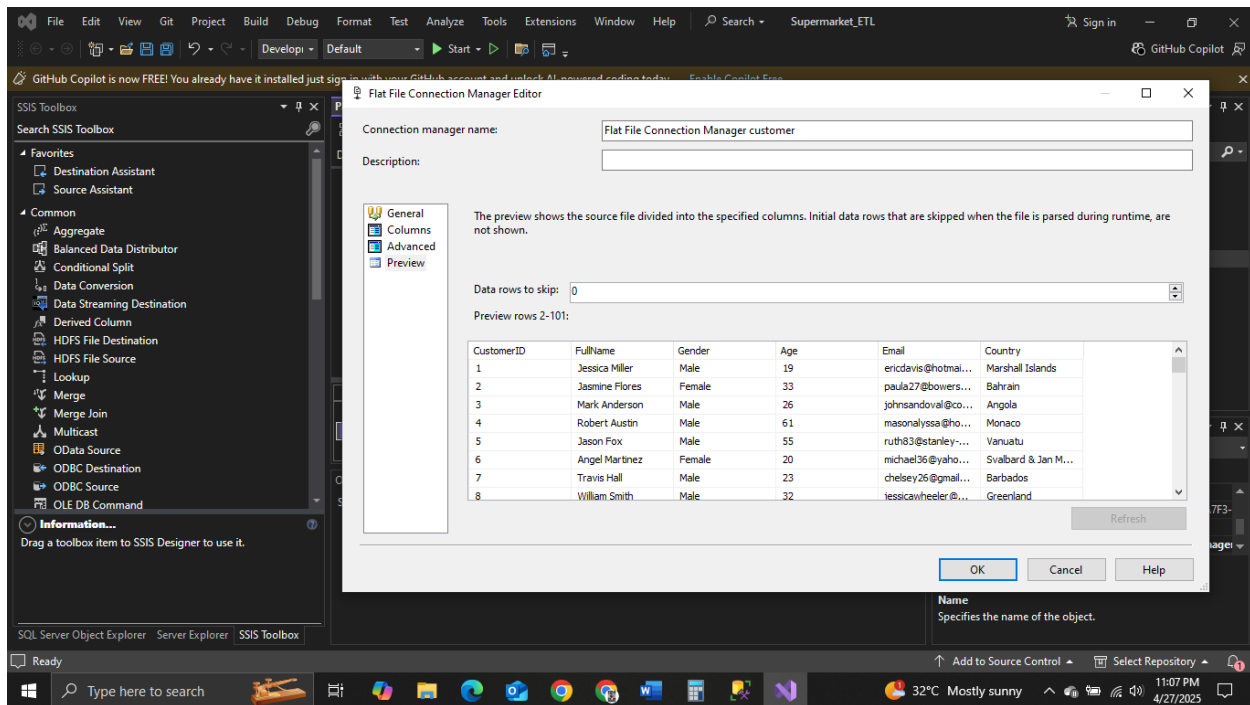
For this project, different types of data sources were set up to simulate the real world. Different types of files were used and were associated with the ETL process through proper connection managers in SQL Server Integration Services.

The table details are as follows:

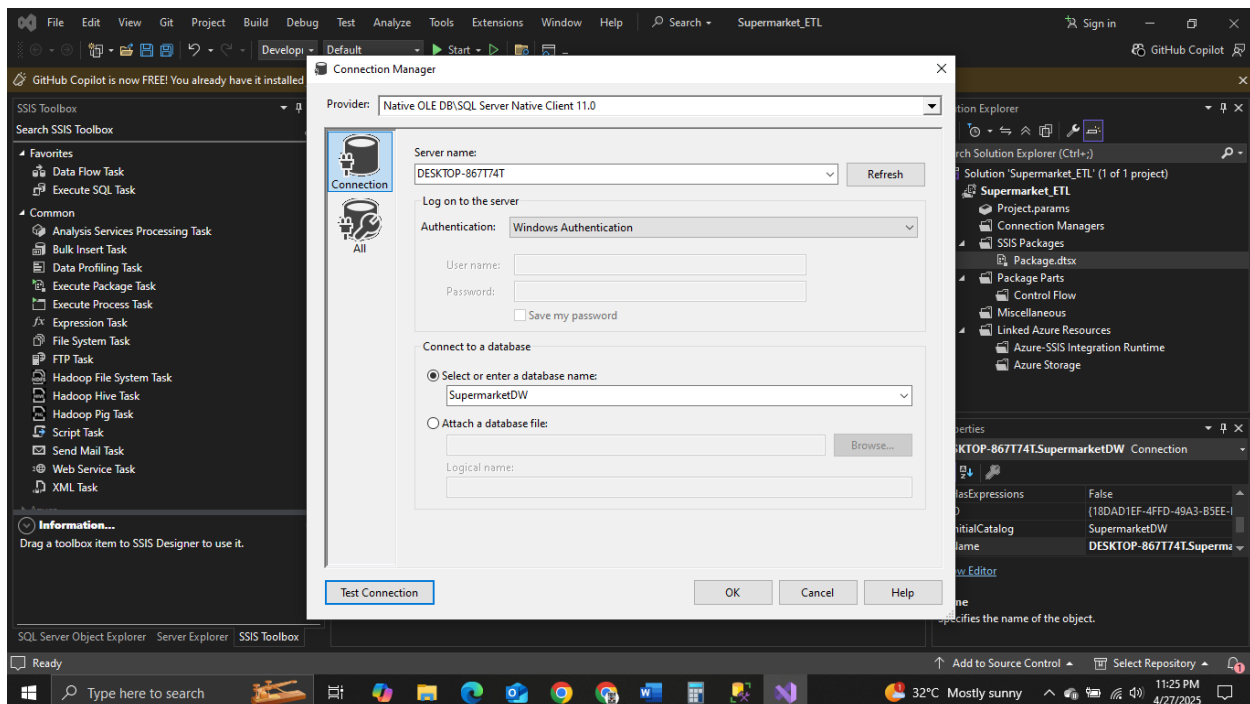
- Customer –
 - Format – Flat File(.csv)
 - Contains customer information such as CustomerID, Full Name, Gender, Email, Age, and Country
- Product –
 - Format – Excel File(.xlsx)
 - contains all details regarding ProductID, ProductName, Category, and UnitPrice..
- Transactions
 - Format - SQL Server Table
 - contains all details regarding TransactionID, ProductID, CustomerID, Quantity, and TransactionDate
- Complete Times
 - Format – Flat File(.csv)
- Used to update completion timestamps for each transaction

Each source was connected to SQL Server Integration Services by appropriate settings using ,

- Using first row as column names
- Verified data types to ensure compatibility

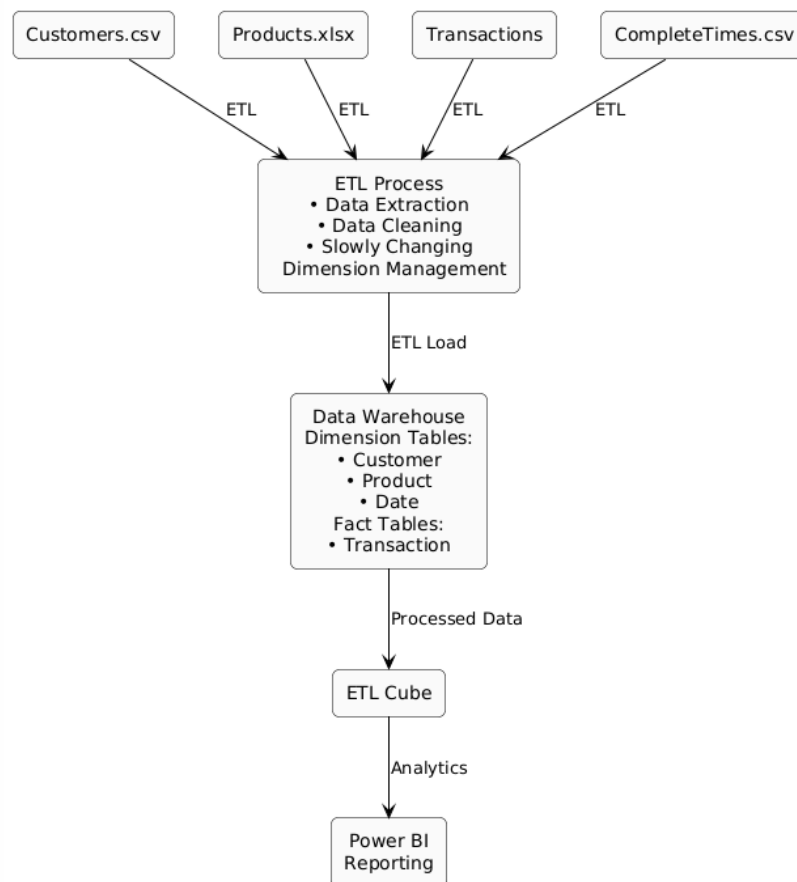


Excel Connection Manager Setup for Products



- Flat File Source Setup for Complete Times

Solution Architecture Design



The architecture integrates multiple data sources providing raw operational data:

- Customers.csv – Stores customer-related information.
- Products.xlsx – Stores product master data.
- Transactions – Stores detailed sales or transactional data.
- CompleteTimes.csv – Stores timestamps and time-based information.
- The data sources may be in diverse formats (CSV, Excel, databases).

The center of the system is the ETL process (Extract, Transform, Load), created using SQL Server Integration Services (SSIS). It performs:

- 1.Data Extraction – Pulls data from source files.
- 2.Data Cleaning – Removes inconsistencies, nulls, duplicates, etc.
- 3.Slowly Changing Dimension (SCD) Management – Handles dimension updates without the loss of history.
- 4.Transformation – Converts data into an appropriate form for analytics.
- 5.Cleaned and transformed data are loaded into the data warehouse.

The core data repository for analysis consists of:

- Dimension Tables:
- Customer
- Product
- Date

Fact Table:

- Transaction

These tables are star schema-based, supporting efficient querying and reporting.

4. ETL Cube

Data from the data warehouse is fed into an ETL cube (OLAP Cube) that pre-aggregates and structures data on multiple dimensions for:

- Faster slicing/dicing
- Drill-down and roll-up operations
- Multidimensional analysis

5. Power BI Reporting

- The last stage links the ETL cube to Power BI, supporting:
- Interactive dashboards
- Visual analytics
- Business insights for decision-makers

6.This architecture guarantees:

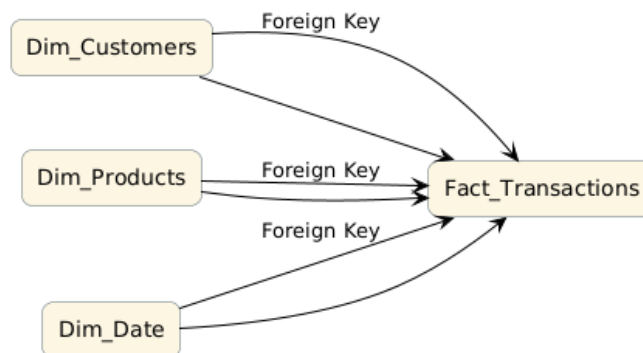
- Scalable, clean, and historical data storage
- Routine reporting
- Real-time business intelligence capabilities

The following diagram shows the structure



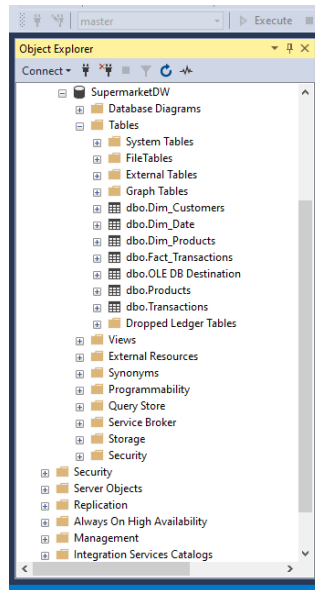
Data Warehouse Design and Development

Supermarket data warehouse is developed according to a Star Schema dimensional model. The schema consists of three-dimension tables and a single fact table, which makes it flexible and efficient for business analysis.



The tables included,

- **Dim_Customers** - Stores customer information
- **Dim_Products** - Stores product details
- **Dim_Date** - Stores detailed date information
- **Fact_Transactions** - Stores transactional information



List of tables inside Supermarket DW

CustomerKey	CustomerID	FullName	Gender	Age	Email	Country	StartDate	EndDate	IsCurrent
1	1	Jessica Miller	Male	19	ericdavis@hotmail.com	Marshall Islands	NULL	NULL	NULL
2	2	Jasmine Flores	Female	33	paula27@browsers.net	Bahrain	NULL	NULL	NULL
3	3	Mark Anderson	Male	26	johnsandoval@cooke-davies.com	Angola	NULL	NULL	NULL
4	4	Robert Austin	Male	61	masonalyse@hotmail.com	Monaco	NULL	NULL	NULL
5	5	Jason Fox	Male	55	ruth93@stanley-mckinney.com	Vanuatu	NULL	NULL	NULL
6	6	Angel Martinez	Female	20	michael36@yahoo.com	Svalbard & Jan Mayen Islands	NULL	NULL	NULL
7	7	Travis Hall	Male	23	chelsey26@gmail.com	Barbados	NULL	NULL	NULL
8	8	William Smith	Male	32	jessicawheeler@hotmail.com	Greenland	NULL	NULL	NULL
9	9	Shawn Sanchez	Male	53	veronica63@alvarez.com	Maldives	NULL	NULL	NULL
10	10	Adam Armstrong	Male	63	sanchezandrew@hotmail.com	Mauritania	NULL	NULL	NULL

Sample Data from Dim Customers

SQLQuery1.sql - DE...eka Fernando (67)*

```
Use SupermarketDW;  
SELECT TOP 10 * FROM Dim_Products;
```

Object Explorer

- SupermarketDW
 - Database Diagrams
 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Dim_Customers
 - dbo.Dim_Date
 - dbo.Dim_Products
 - dbo.Fact_Transactions
 - dbo.OLE DB Destination
 - dbo.Products
 - dbo.Transactions
 - Dropped Ledger Tables
 - Views
 - External Resources
 - Synonyms
 - Programmability
 - Query Store
 - Service Broker
 - Storage
 - Security
 - Server Objects
 - Replication
 - Always On High Availability
 - Management
 - Integration Services Catalogs

Results

	ProductKey	ProductID	ProductName	Category	UnitPrice
1	1	1	Physical Beverages	Food > Beverages	13.15
2	2	2	Key Beverages	Food > Beverages	1.48
3	3	3	Return Beverages	Food > Beverages	6.23
4	4	4	Stay Beverages	Food > Beverages	5.24
5	5	5	Lot Beverages	Food > Beverages	14.99
6	6	6	Report Beverages	Food > Beverages	13.86
7	7	7	Mouth Beverages	Food > Beverages	17.95
8	8	8	Team Beverages	Food > Beverages	2.65
9	9	9	Threat Beverages	Food > Beverages	9.02
10	10	10	Training Beverages	Food > Beverages	1.57

Query executed successfully.

DESKTOP-867T74T (16.0 RTM) DESKTOP-867T74T\Buwane... SupermarketDW 00:00:00 10 rows

Sample Data from Dim_Products

SQLQuery1.sql - DE...eka Fernando (67)*

```
Use SupermarketDW;  
SELECT TOP 10 * FROM Fact_Transactions;
```

Object Explorer

- SupermarketDW
 - Database Diagrams
 - Tables
 - System Tables
 - FileTables
 - External Tables
 - Graph Tables
 - dbo.Dim_Customers
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 - dbo.Dim_Products
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 - Server Objects
 - Replication
 - Always On High Availability
 - Management
 - Integration Services Catalogs

Results

	TransactionID	CustomerKey	ProductKey	DateKey	Quantity	acon_bn_create_time	acon_bn_complete_time	bn_process_time_hours
1	1	31	11	20240905	7	2025-04-18 16:28:12.910	2024-01-01 11:33:00.000	-11357
2	2	62	14	20240131	7	2025-04-18 16:28:12.910	2024-01-01 09:00:00.000	-11359
3	3	22	25	20240718	1	2025-04-18 16:28:12.910	2024-01-01 10:12:00.000	-11358
4	4	34	30	20240804	5	2025-04-18 16:28:12.910	2024-01-01 10:45:00.000	-11358
5	5	90	47	20241204	9	2025-04-18 16:28:12.910	2024-01-01 09:38:00.000	-11359
6	6	92	32	20240407	3	2025-04-18 16:28:12.910	2024-01-01 09:30:00.000	-11359
7	7	38	14	20241023	1	2025-04-18 16:28:12.910	2024-01-01 10:25:00.000	-11358
8	8	95	35	20240609	1	2025-04-18 16:28:12.910	2024-01-01 13:07:00.000	-11355
9	9	8	4	20240901	10	2025-04-18 16:28:12.910	2024-01-01 13:15:00.000	-11355
10	10	65	34	20240130	3	2025-04-18 16:28:12.910	2024-01-01 13:02:00.000	-11355

Query executed successfully.

DESKTOP-867T74T (16.0 RTM) DESKTOP-867T74T\Buwane... SupermarketDW 00:00:00 10 rows

ETL Development

The ETL (Extract, Transform, Load) process was developed using SQL Server Integration Services. It involves data extractions from various sources, transformed as needed and then loaded into the data warehouse following the star schema model.**Data Sources**

Multiple types of data sources were used:

- **CSV Files:** customers.csv, complete_times.csv
- **Excel File:** products.xlsx
- **SQL Server Table:** transactions

SSIS Tasks and Transformations

- Flat File Source - To extract customer and complete time data from CSV files
- Excel Source - To extract product data from an excel sheet
- OLE DB Source - To extract transaction data from SQL Server table
- Data Conversion - To convert data types
- Derived Column - To create new calculated fields such as accm_txn_create_time
- OLE DB Destination - To load the cleaned and transformed data into the data warehouse tables
- OLE DB Command - To update completion timestamps
- Lookup Transformations - To map CustomerID, ProductID, and TransactionDate to their related dimension keys

Transformations Types

Data Type Conversion - Controlled inconsistencies between warehouse schema and source data types.

Lookup - Mapped Transaction foreign keys

Derived Column - Added system timestamps and duplicated fields for updates

ETL Development – Accumulating Fact Tables

1. Fact Table Extension

In the Fact_Transactions table, the table was extended by adding these columns

- accm_txn_create_time - Indicates the time when the transaction record was created (datetime)
- accm_txn_complete_time - Indicates the actual time when the transaction was completed.(datetime)
- txn_process_time_hours - The difference between completion time and creation time in hours. (int)

2. Setting accm_txn_create_time

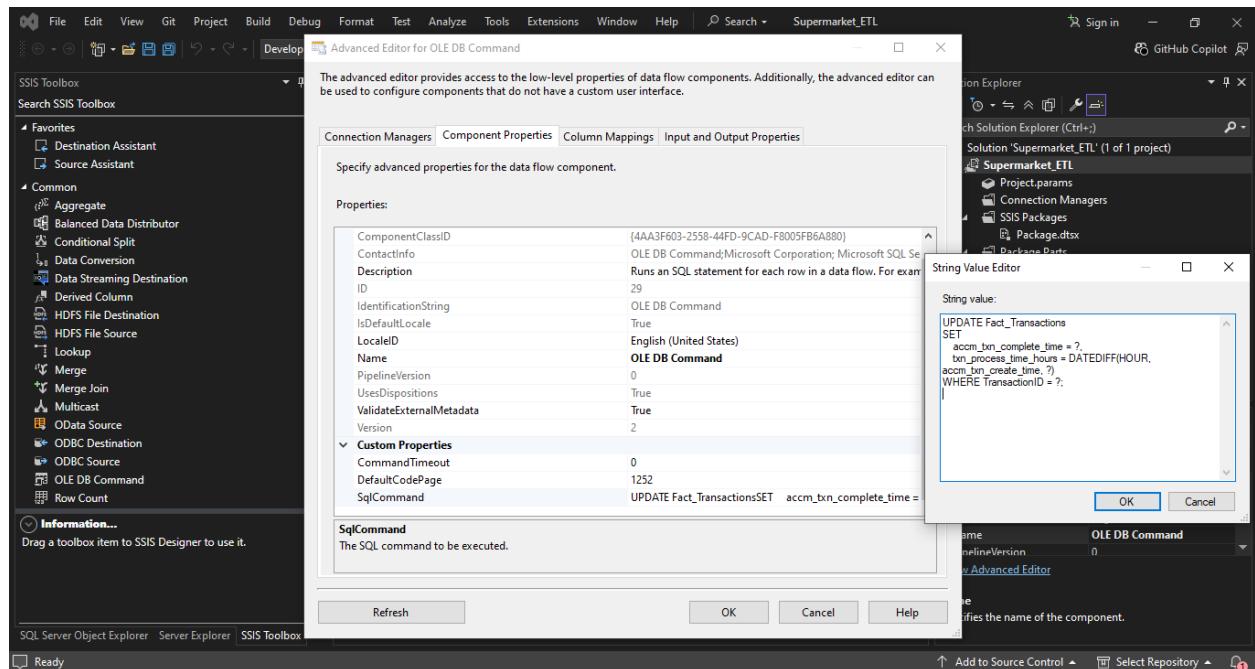
During the initial ETL load of transactions, a **Derived Column** in Server Integration Services was used to set accm_txn_create_time by using GETDATE() function to get the current date and time.

3. In the completion of dataset, a separate dataset called complete_times.csv was created, which containing **TransactionID** and **accm_txn_complete_time**.

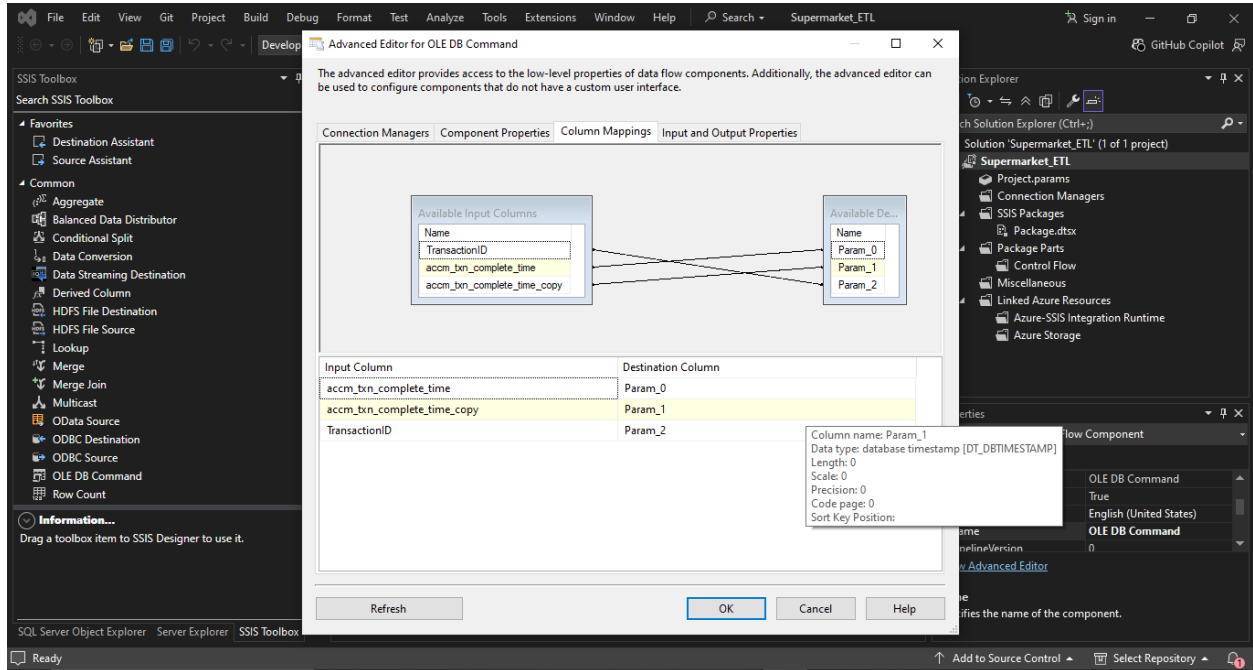
4. Updating Fact Table

- Standalone SSIS Data Flow has been developed to update the Fact Table as given below::
 - Step 1 - Flat File Source reads complete_times.csv
 - Step 2 - Derived Column duplicates accm_txn_complete_time to use for parameter Mapping
 - accm_txn_complete_time for parameter mapping
 - **Step 3** - OLE DB Command updates Fact_Transactions setting in
 - **accm_txn_complete_time**
 - **txn_process_time_hours = DATEDIFF(HOUR, accm_txn_create_time, accm_txn_complete_time)**

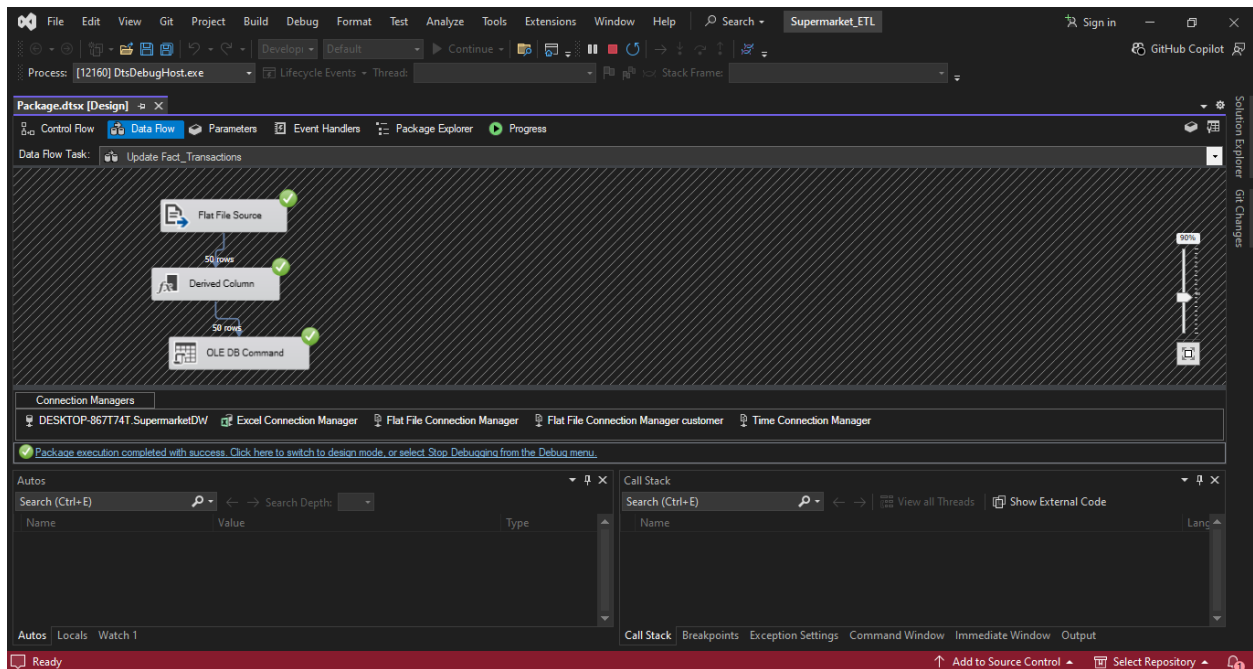
5. accm_txn_complete_time :



Display how the Mapping is done



Data-Flow Of Fact Transaction



Data Display in the Accumilated fact table

The screenshot displays the Microsoft SQL Server Management Studio interface. The title bar indicates the file is 'SQLQuery1.sql' and the server is 'DESKTOP-867T74T\SUPERMARKETDW'. The 'Object Explorer' on the left shows the server structure, including 'Databases', 'Security', 'Server Objects', 'Replication', 'Always On High Availability', 'Management', 'Integration Services Catalogs', 'SQL Server Agent (Agent XPs disabled)', and 'XEvent Profiler'. The 'Query Editor' in the center contains the following SQL query:

```
use SupermarketDW;

SELECT TOP 10
    TransactionID,
    acm_txn_create_time,
    acm_txn_complete_time,
    txn_process_time_hours
FROM Fact_Transactions
WHERE acm_txn_complete_time IS NOT NULL;
```

The 'Results' pane at the bottom shows the output of the query, which consists of 10 rows. The columns are 'TransactionID', 'acm_txn_create_time', 'acm_txn_complete_time', and 'txn_process_time_hours'. The data is as follows:

TransactionID	acm_txn_create_time	acm_txn_complete_time	txn_process_time_hours
1	2025-04-18 16:28:12.910	2024-01-01 11:33:00.000	-11357
2	2025-04-18 16:28:12.910	2024-01-01 09:00:00.000	-11359
3	2025-04-18 16:28:12.910	2024-01-01 10:12:00.000	-11358
4	2025-04-18 16:28:12.910	2024-01-01 10:45:00.000	-11358
5	2025-04-18 16:28:12.910	2024-01-01 09:38:00.000	-11359
6	2025-04-18 16:28:12.910	2024-01-01 09:30:00.000	-11359
7	2025-04-18 16:28:12.910	2024-01-01 10:25:00.000	-11358
8	2025-04-18 16:28:12.910	2024-01-01 13:07:00.000	-11355
9	2025-04-18 16:28:12.910	2024-01-01 13:15:00.000	-11355
10	2025-04-18 16:28:12.910	2024-01-01 13:02:00.000	-11355

The status bar at the bottom indicates 'Query executed successfully.' and '10 rows'.

Summery of the project

This project successfully illustrates end-to-end development of a retail sales data warehouse solution using Microsoft SQL Server and SQL Server Integration Services (SSIS).

Key accomplishments are:

- Preparation of a reasonable OLTP dataset for customer, product, and transaction data for one year.
- Dimension tables and a fact table in data warehouse using the Star Schema model.
- Implementing a complete ETL pipeline using SQL Server Integration Services, involving data extraction from multiple sources and also, data transformations, lookup functions, and loading procedures.
- Implementing an update pipeline for transaction completion times and computing process durations to manage the generation and updating of an accumulating fact table.
- Maintaining the data warehouse integrity by executing validation queries and correct loading and updating of records.

Overall, the project demonstrated important skills in data modeling, ETL development, and application of data warehousing concepts in practice.